

Robotics-for-Cacao Barter

Follow-up action item · TrueSight DAO / Agroverse · 2026-08-06

Context. Three field priorities: harvesting cacao, cutting off witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*), and cacao roasting flavor-profile control. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, <https://youtu.be/hg8qYrjyYCU>). This follow-up turns that introduction into a concrete pilot.

1 · The problem on the farm

- **Witches' broom** is the single highest-leverage canopy disease in Bahia cacao: infected brooms must be pruned out **and removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees.
- **Labor shortage:** pruning and selective pod harvest are manual, seasonal, and hard to staff.
- **Cash constraint:** smallholders have little cash but hold **cacao** — the natural payment asset.

2 · Working prototype reference — Frasky (IIT)

Video (Reuters): <https://youtu.be/hg8qYrjyYCU> — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).
- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.
- Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

Proof point: a permanent tree/vine crop's canopy work can be robotized. Cacao is the same category — this is the working prototype we point partners and farmers to.

Frasky capability (grape)	Cacao adaptation	Feasibility
Cluster monitoring via camera	Broom / pod detection via vision	■ tractable
Manipulate plants (robotic arm)	Broom pruning with cutter arm	■ tractable
Targeted spray treatments	Targeted biocontrol on brooms	■ tractable
—	Selective pod <i>harvest</i> (varied heights: trunk + branches)	■■■ harder R&D

3 · The proposal (barter model)

Offer robotics services to Bahia/Pará farmers **in exchange for cacao as payment:**

Role	Who
Robot design & build	Robotics design expert (Gianluca)
Farmer network & deployment	Agroverse — Bahia operations (Ilhéus)
Market for cacao received	TrueSight DAO / Agroverse (US import live)

Honest MVP scoping. Broom *detection + pruning* (or monitoring + targeted treatment, à la Frasky) is tractable now. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable.

4 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (<https://youtu.be/hg8qYrjyYCU>) as the reference prototype; define MVP build, season timing, and cost basis.
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL.
3. **2–3 test farms** in the Bahia (Ilhéus) network.
4. **Barter accounting** — value cacao received at market BRL/USD; record as INVENTORY MOVEMENT + purchase into the Main Ledger (see SUPPLY_CHAIN_AND_FREIGHTING.md).
5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved. No scaling on unmeasured evidence.

5 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility questions) and the PT-BR farmer pitch.
- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

6 · Tie-back to mission

Every broom pruned and every robot-assisted harvest keeps more trees producing and more **Amazon rainforest standing** — the DAO's 10,000-hectare regeneration goal is served by making cacao farming viable without clearing new land.

Filed in [agentic_ai_context/OPEN_FOLLOWUPS.md](#) (PR #733) · nothing sent to any party yet — awaiting governor approval of outreach copy.